

put ③ in ① then

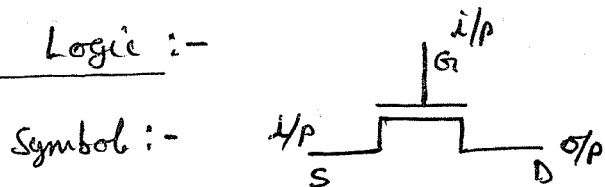
$$I_{max} = \frac{C_{ox} W \mu_n}{L} \left[(V_{GS} - V_{th}) \cdot (V_{GS} - V_{th}) - \frac{(V_{GS} - V_{th})^2}{2} \right]$$

$$\boxed{I_{max} = \frac{C_{ox} W \mu_n}{2L} [V_{GS} - V_{th}]^2} \quad \text{--- ④}$$

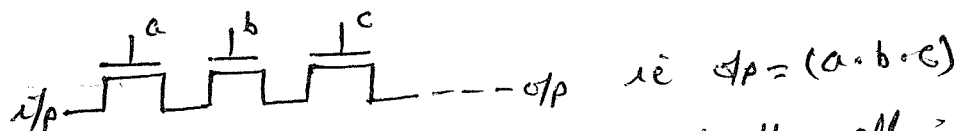
i.e maximum current in saturation region

Logic families

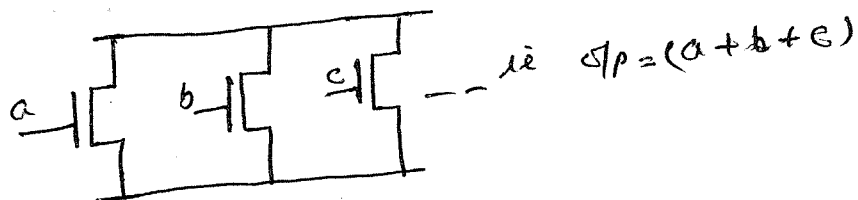
① NMOS Logic :-



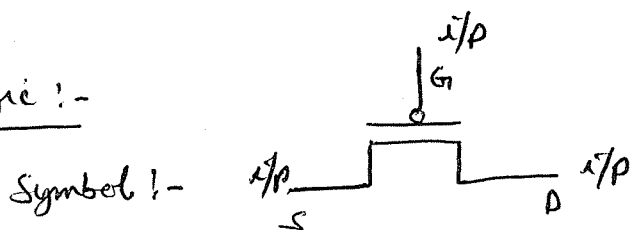
- for ON $\Rightarrow$ (i) if Gate = logic '1', then o/p connected to i/p.
 for OFF $\Rightarrow$ (ii) if Gate = logic '0', then o/p ^{not} connected to i/p.
 (iii) when nmos connected in series then all are multiply.



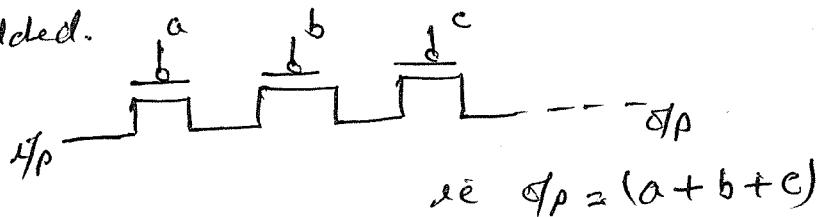
- (iv) when nmos connected in parallel then all are added.



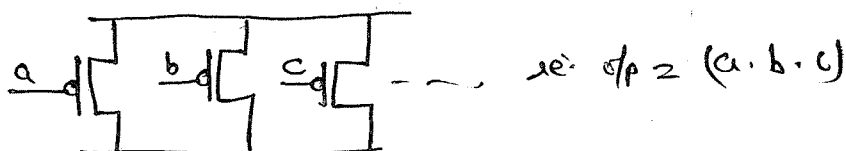
② PMOS Logic :-



- for ON $\Rightarrow$ (i) if Gate = logic '0', then o/p connected to i/p.
 for OFF $\Rightarrow$ (ii) if Gate = logic '1', then o/p ^{not} connected to i/p.
 (iii) when pmos connected in series then all are added.

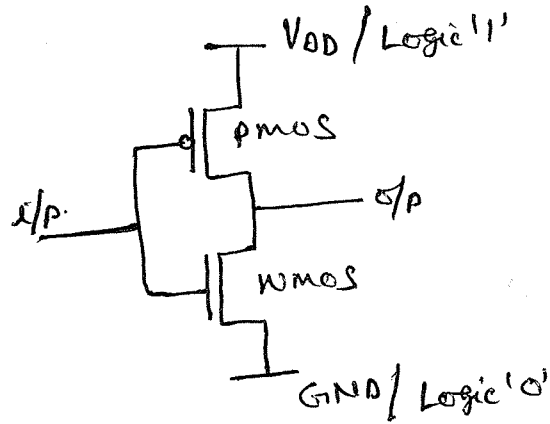


- (iv) when pmos connected in parallel then all are multiplied.



③ CMOS Logic:- (Complementary MOS Logic)

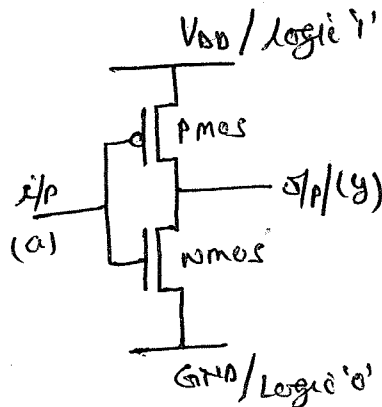
when one pmos connected with nmos in series.



- (i) when $\bar{x}/p = '0'$ then pmos = ON & nmos = OFF, then logic '1' / V_{DD} connected to o/p.
- (ii) when $\bar{x}/p = '1'$ then pmos = OFF & nmos = ON, then logic '0' / GND connected to o/p.

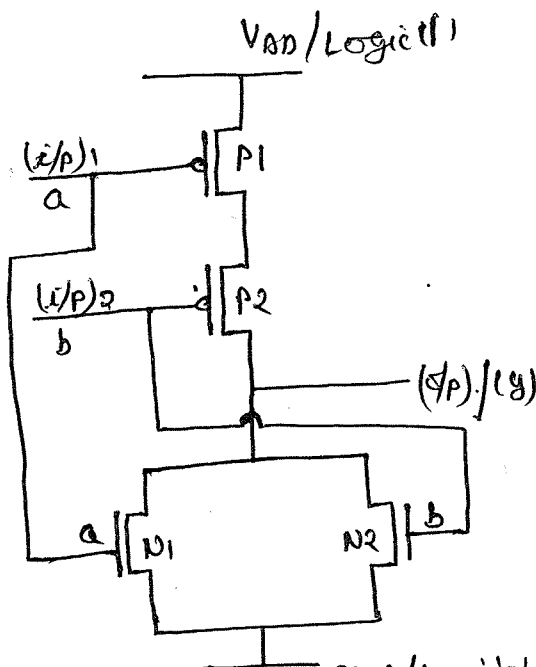
example:-

(i)



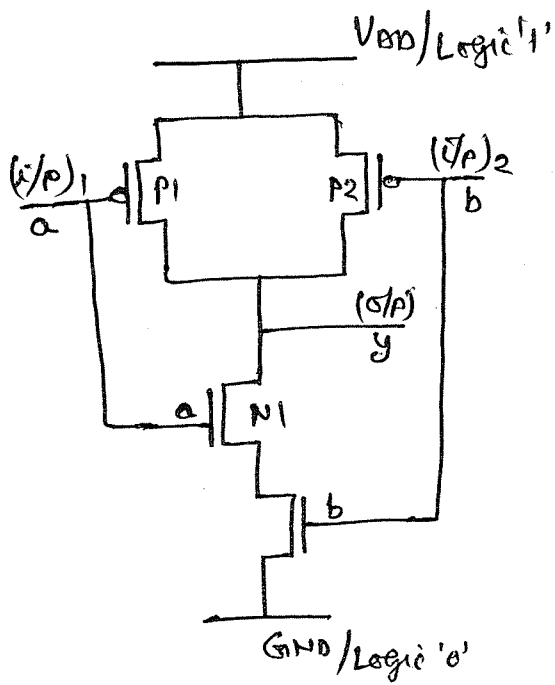
a	y (NOT gate)
0	1
1	0

(ii)



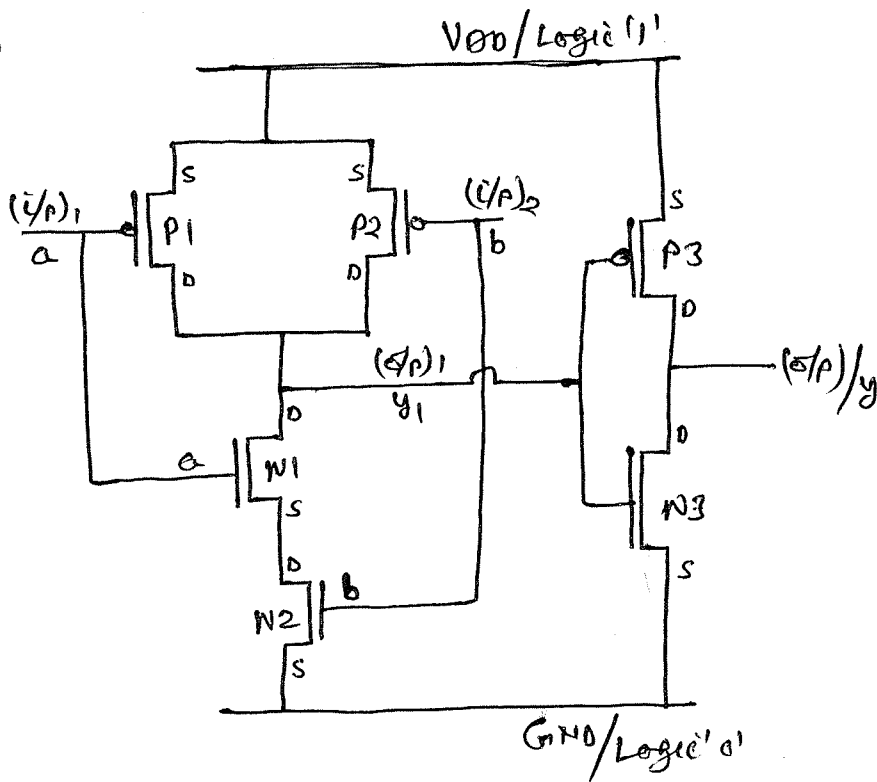
a	b	y (NOR gate)
0	0	1
0	1	0
1	0	0
1	1	0

(iii)



a	b	y (NAND gate)
0	0	1
0	1	1
1	0	1
1	1	0

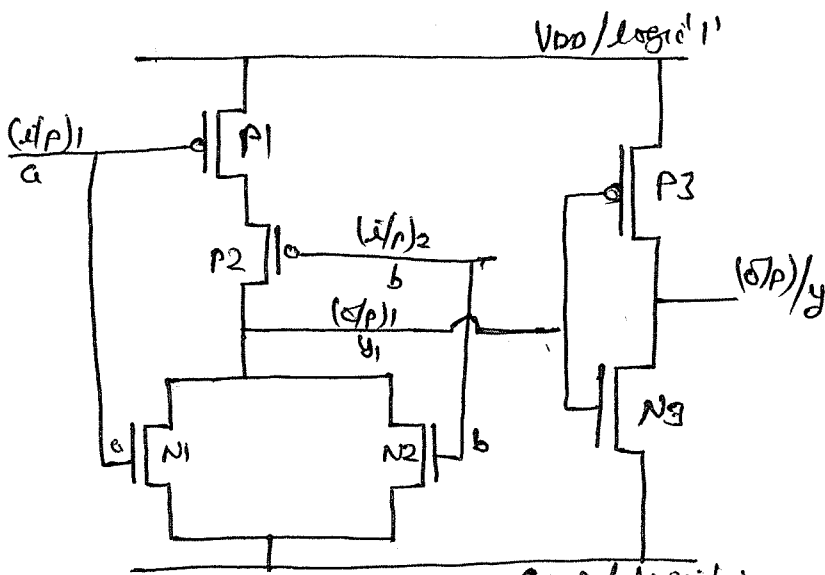
(iv)



a	b	y ₁	y
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

(AND gate)

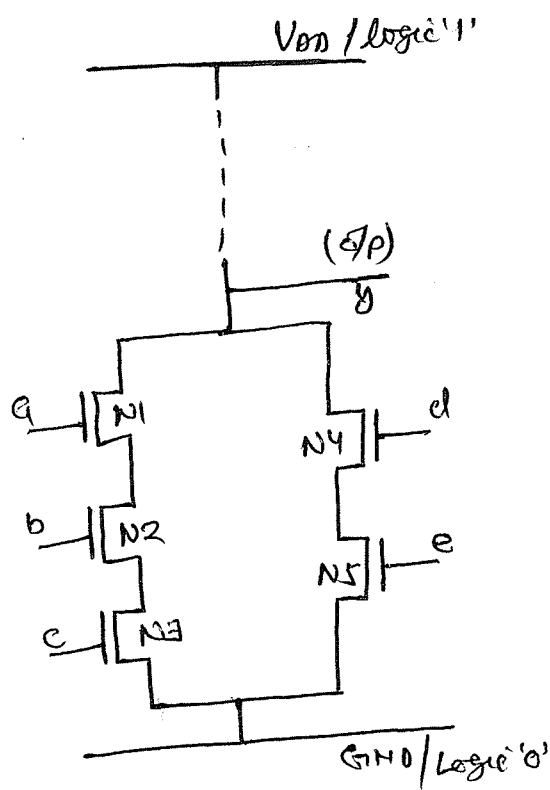
(v)



a	b	y ₁	y
0	0	1	0
0	1	0	1
1	0	0	1
1	1	0	1

(OR gate)

(vi)



first write function using
Nmos combination only. the
take complement always.
$$y = (a \cdot b \cdot c) + (d \cdot e)$$

In diagram,
Nmos = series (a, b, c) then
Pmos = parallel (a, b, c)
Nmos = series (d, e) then
Pmos = parallel (d, e)

ie final logic diagram,

